

Total No. of Questions : 6]

SEAT No. :

P5

[Total No. of Pages : 2

APR.-17/BE/Insem.-5

B.E. (Civil)

HYDROPOWER ENGINEERING

(2012 Pattern) (Elective - III(c))

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.*
- 2) Figures to the right side indicate full marks.*
- 3) Use of Calculator is allowed.*
- 4) Assume Suitable data if necessary.*

- Q1)** a) Explain the present status of hydropower generation in India. [5]
b) What are the effects of climate change and green house effects on hydropower generation and its development? [5]

OR

- Q2)** a) Explain various trends in energy use patterns in India. [5]
b) Distinguish between thermal power and hydro power. [5]

- Q3)** a) Draw a schematic layout of hydro electric power plant. Enlist the functions of transformer, and penstock. [4]
b) What a detail note on: run of river plant with and without pondage.[6]

OR

- Q4)** a) What is surge tank? What are the different types of surge tank? Explain with the help of neat sketches. What are the functions of surge tanks?[6]
b) What are the criteria for the selection of site for hydro electric power plant? [4]

P.T.O.

- Q5) a)** A run of river hydro electric power station is proposed across a river at a site where a net head of 15m is available on the turbine. The river carries a sustained minimum flow of 30 cumec with the load factor of 70%. Plant efficiency is 50%. Determine the maximum generating capacity of the generator to be installed at the power house. If the daily load pattern includes 21 hours average load and 3 hours of peak load, determine the volume of pondage to be provided to supply the daily demand. [5]
- b) Define diversity factor, demand factor, connected load, Firm power and secondary power. [5]

OR

- Q6) a)** Show that capacity factor is equal to product of load factor and utilization factor. [4]
- b) A load on hydel plant varies from minimum of 10000 kW to a maximum of 50000 kW. Two turbo generators of capacities 30000 kW each have been installed. Calculate [6]
- i) Total installed capacity of the plant
 - ii) plant factor
 - iii) maximum demand
 - iv) load factor
 - v) utilization factor

